OXFORD AQA
INTERNATIONAL QUALIFICATIONS

Please write clearly in block capitals.

Centre number | | | | | | Candidate number | | | | |

Surname ____________________

Forename(s) ____________________

Candidate signature ____________________
I declare this is my own work.

INTERNATIONAL GCSE
BIOLOGY
Paper 1

Wednesday 25 October 2023 07:00 GMT Time allowed: 1 hour 30 minutes

Materials
For this paper you must have:
- a ruler with millimetre measurements
- a scientific calculator.

Instructions
- Use black ink or black ball-point pen.
- Fill in the boxes at the top of this page.
- Answer **all** questions in the spaces provided.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information
- There are 90 marks available on this paper.
- The marks for questions are shown in brackets.
- You are expected to use a calculator where appropriate.
- You are reminded of the need for good English and clear presentation in your answers.

Advice
In all calculations, show clearly how you work out your answer.

For Examiner's Use	
Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	

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Answer **all** questions in the spaces provided.

0 1 All plants and animals have similar characteristics to their parents.

0 1 . 1 What is the basic unit of information that is passed on from parent to offspring?

[1 mark]

Tick (✓) **one** box.

A gene ☐

A hormone ☐

A ribosome ☐

0 1 . 2 Which cells carry units of information about inherited characteristics from parent to offspring?

[1 mark]

Tick (✓) **one** box.

Blood cells ☐

Liver cells ☐

Sex cells ☐

Skin cells ☐

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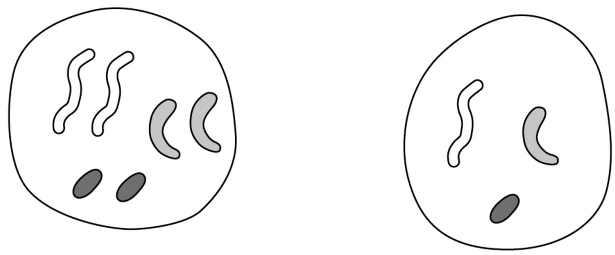
0 1 . 3 The nucleus of a cell contains chromosomes which carry the information about inherited characteristics.

Figure 1 shows the chromosomes in the nucleus of two different cells from a muntjac deer.

Figure 1



Which cell nucleus is from a skin cell and which cell nucleus is from a sex cell?

Give a reason for each answer.

[2 marks]

Cell **A** nucleus ____

____

Cell **B** nucleus ____

____

0 1 . 4 How many chromosomes are there in a **human** skin cell?

[1 mark]

Tick (✓) **one** box.

2 ☐ 16 ☐ 23 ☐ 46 ☐

Question 1 continues on the next page

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0 1 . 5 The human sex chromosomes are X and Y.

Give the genotype of a male and of a female.

[2 marks]

Male ____________________________

Female ____________________________

0 1 . 6 Alleles control the development of characteristics.

If a person has two **different** alleles for a characteristic they are heterozygous.

Complete the sentence.

[1 mark]

If a person has two alleles which are the **same** for a characteristic

they are ____________________________.

Do not write outside the box

In one species of mouse brown fur is caused by a dominant allele, **B**.

The recessive allele for white fur is **b**.

0 1 . 7 Complete the genetic diagram in **Figure 2**.

[2 marks]

Figure 2

		Female	
		B	**b**
Male	**B**	BB	
	b		

0 1 . 8 Draw a ring around each of the offspring genotypes in **Figure 2** that will result in a mouse with brown fur.

[1 mark]

0 1 . 9 What is the ratio of brown fur to white fur in the offspring in **Figure 2**?

[1 mark]

Tick (✓) **one** box.

1 : 3 ☐ 1 : 1 ☐ 3 : 1 ☐ 4 : 1 ☐

12

Turn over for the next question

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There are no questions printed on this page

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ANSWER IN THE SPACES PROVIDED

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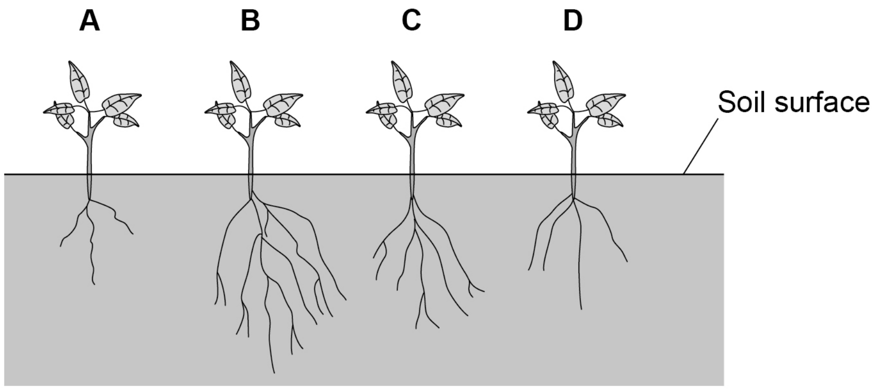
0 2 Plants absorb water through their roots and lose water through their leaves.

0 2 . 1 **Figure 3** shows four plants.

Figure 3



Which plant will absorb the most water in one hour?

[1 mark]

Tick (✓) **one** box.

A ☐ **B** ☐ **C** ☐ **D** ☐

Question 2 continues on the next page

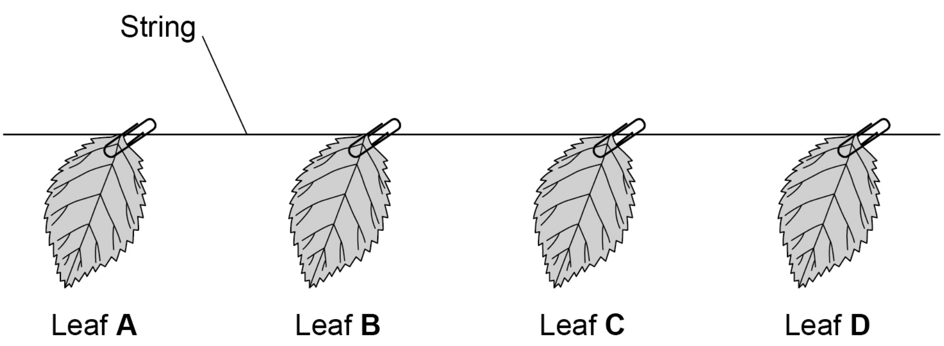
A student measured the water loss from leaves.

This is the method used.

1. Measure the surface area of each leaf using squared graph paper.
2. Measure the thickness of each leaf.
3. Calculate the volume of each leaf.
4. Measure the mass of each leaf.
5. Hang the leaves on a string in a laboratory for four hours.
6. Measure the mass of each leaf again.
7. Calculate the decrease in mass of each leaf.

Figure 4 shows the leaves.

Figure 4



0 2 . 2 Complete the sentence.

[1 mark]

Choose the answer from the box.

balance	**pipette**	**ruler**	**thermometer**

The instrument used to measure mass is called a ____________________________.

Do not write outside the box

Table 1 shows the results.

The decrease in mass is due to water loss.

Table 1

Leaf	**A**	**B**	**C**	**D**
Surface Area : Volume (SA : Vol)	20 : 1	32 : 1	43 : 1	56 : 1
Decrease in mass in g	0.055	0.076	0.101	0.125

0 2 . 3 What conclusion can the student make about surface area to volume ratio and water loss?

Complete the sentence.

[1 mark]

Choose your answer from the box.

increases	**stays the same**	**decreases**

As surface area to volume ratio increases, water loss ____________________________.

Question 2 continues on the next page

Turn over ►

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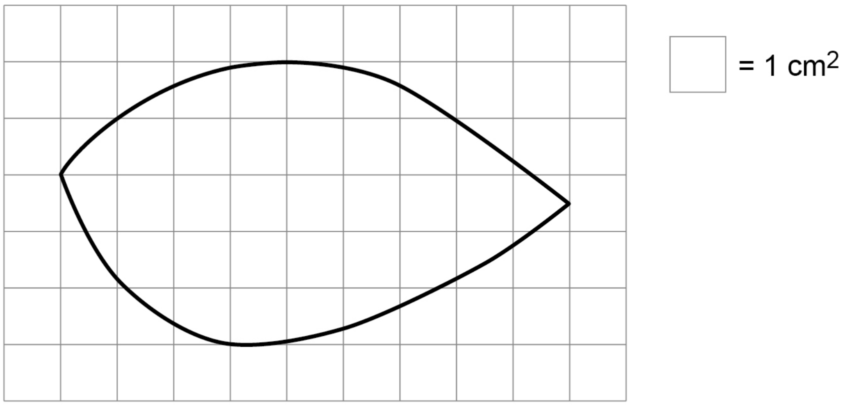
0 2 . 4 The student repeated the investigation using two more leaves, **E** and **F**.

Figure 5 shows an outline of leaf **E** on squared graph paper.

Figure 5



Determine the surface area of the leaf outline in **Figure 5**.

[1 mark]

___

Surface area = ______________ cm^2

0 2 . 5 Leaf **F** had:

- a surface area of 11.0 cm^2 calculated using squared graph paper
- a thickness of 0.12 cm.

Calculate the volume of leaf **F**.

[2 marks]

Use the equation:

volume = surface area × thickness

___

___

Volume = ______________ cm^3

Do not write outside the box

0 2 . 6 The student wanted to calculate the **total** surface area of each leaf.

The student had to multiply the surface area found using squared graph paper by 2.

Give the reason why the student had to multiply by 2.

[1 mark]

____

0 2 . 7 Calculate the surface area to volume ratio (SA : Vol) for leaf **F**.

Give your answer to the nearest whole number.

[3 marks]

____

____

____

____

____

SA : Vol = ____________________ : 1

Question 2 continues on the next page

Do not write outside the box

0 2 . 8 The student repeated the experiment outside on two days.

Figure 6 shows the environmental conditions on the two days.

Figure 6

Day 1	**Day 2**
Light intensity = 180 lux	Light intensity = 190 lux
Temperature = 25 °C	Temperature = 15 °C
Wind speed = 26 km/h	Wind speed = 3 km/h
Humidity = 56%	Humidity = 55%

Give **two** reasons why the leaves lose water faster on **Day 1** than on **Day 2**.

[2 marks]

1 ____

2 ____

0 2 . 9 Water is lost from the leaves through the stomata.

Describe how the loss of water from leaves is controlled.

[2 marks]

____

____

____

____

____

14

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0 3 Homeostasis controls some conditions in the body.

0 3 . 1 Which condition is controlled by homeostasis?

[1 mark]

Tick (✓) **one** box.

Body temperature ☐

Eyesight ☐

Growth rate ☐

Hair colour ☐

The water content and ion content in the body are also controlled by homeostasis.

0 3 . 2 How does homeostasis control the water content in the body?

[1 mark]

Tick (✓) **one** box.

By gaining water in the diet ☐

By losing fixed volumes every hour ☐

By using automatic systems ☐

0 3 . 3 Water leaves the body in sweat.

Give **one other** way that water leaves the body.

[1 mark]

____

Do not write outside the box

0 3 . 4 Name **two** substances that are lost from the body in sweat.

Do **not** refer to water.

[2 marks]

1 ____

2 ____

0 3 . 5 The kidneys help to:

- remove unwanted substances from the body
- keep substances that are needed in the body.

Describe how a healthy kidney functions.

[4 marks]

Do not write outside the box

0 3 . 6 The liver also helps to remove unwanted substances from the body.

Describe **two** ways in which the liver helps to remove unwanted substances.

[2 marks]

1 ____

____

2 ____

____

11

Turn over for the next question

0 4 Students investigated the effect of pH on the activity of a protease enzyme.

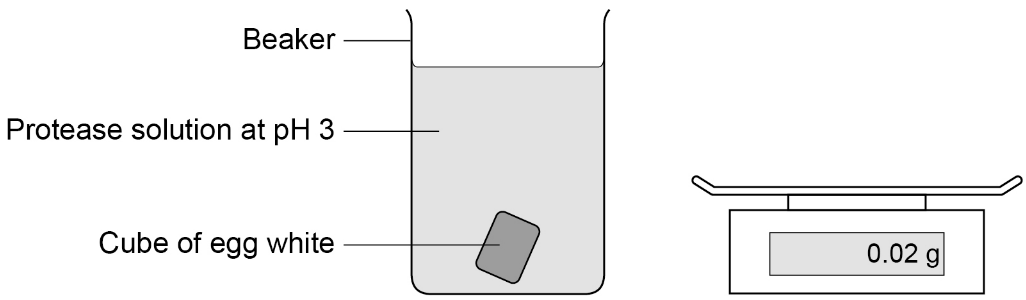
The protein used in the investigation was albumen in egg white.

This is the method used.

1. Prepare 2 cm cubes of solid egg white.
2. Record the mass of one egg white cube.
3. Place the cube in a beaker with 50 cm^3 of protease solution at pH 3.
4. After 3 hours remove the cube, rinse in water and blot dry.
5. Record the new mass of the egg white cube.
6. Calculate the decrease in mass of the egg white cube.
7. Repeat steps 2 to 6 two more times.
8. Repeat steps 2 to 7 with pH 5, pH 7 and pH 9 solutions.

Figure 7 shows the apparatus.

Figure 7



0 4 . 1 Give **one** control variable the students should have used.

[1 mark]

____

Table 2 shows the results.

Table 2

pH	Decrease in mass of egg white cube in g			
	Test 1	**Test 2**	**Test 3**	**Mean**
3	0.89	0.93	0.94	0.92
5	1.25	1.17	1.21	1.21
7	0.57	0.55	0.53	0.55
9	0.26	0.31	0.27	**X**

0 4 . 2 Calculate value **X** in **Table 2**.

[2 marks]

Value **X** = ______________________ g

0 4 . 3 What conclusion can the students make about the activity of the protease enzyme?

Use **Table 2**.

[1 mark]

0 4 . 4 Describe **one** improvement to the method that would improve the validity of the conclusion.

Do **not** refer to repeating the investigation.

[1 mark]

Do not write outside the box

The students noticed that their electronic balance was reading 0.02 g when there was nothing on the pan.

0 4 . 5 What type of error is this?

[1 mark]

____

0 4 . 6 How should the students correct the error with the electronic balance?

[1 mark]

____

____

0 4 . 7 After 3 hours there will be a new type of molecule present in the beakers.

Name the new type of molecule.

Give the reason why it has appeared.

[2 marks]

Type of molecule ____

Reason ____

____

Do not write outside the box

0 4 . 8 Neurones contain proteins that help them to function effectively.

Condition **X** is a medical condition where proteins in neurones are broken down.

The damaged neurones disrupt nervous impulses.

A person with condition **X** cannot coordinate their movements to pick up an object.

Describe how damage to each of the three types of neurone would disrupt different impulses in the nervous system.

You should refer to the names of each type of neurone.

[3 marks]

12

Turn over for the next question

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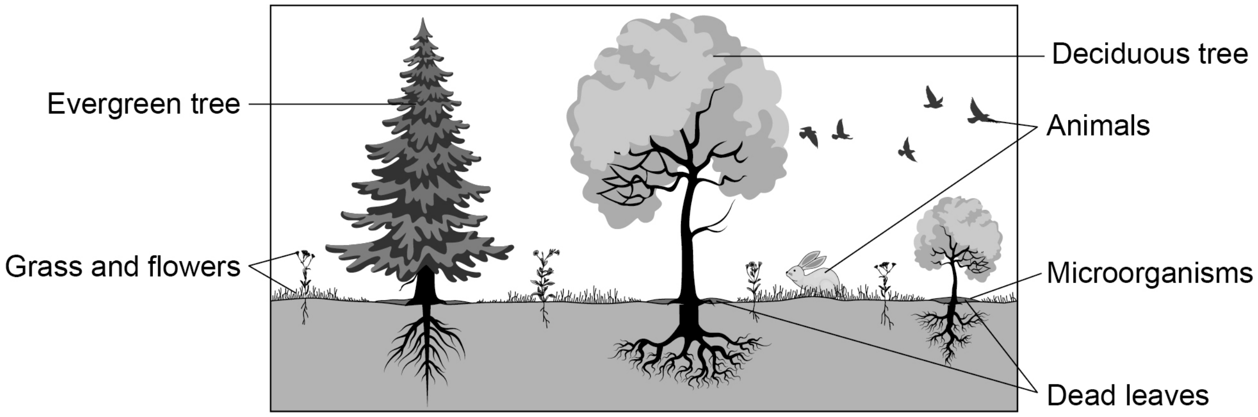
0 5 Substances are cycled through communities.

Figure 8 shows part of a forest community.

Figure 8



0 5 . 1 What is the forest community?

[1 mark]

Tick (✓) **one** box.

All the changes in the forest in one year ☐

All the light, water and oxygen in the forest ☐

All the microorganisms and animals in the forest ☐

All the populations of living things in the forest ☐

0 5 . 2 The forest community is a stable community.

What happens in a stable community?

[1 mark]

Tick (✓) **one** box.

Processes that remove materials are balanced by processes that return materials. ☐

Processes that remove waste materials are able to slowly clean up the environment. ☐

Processes that return materials are able to capture carbon and reduce global warming. ☐

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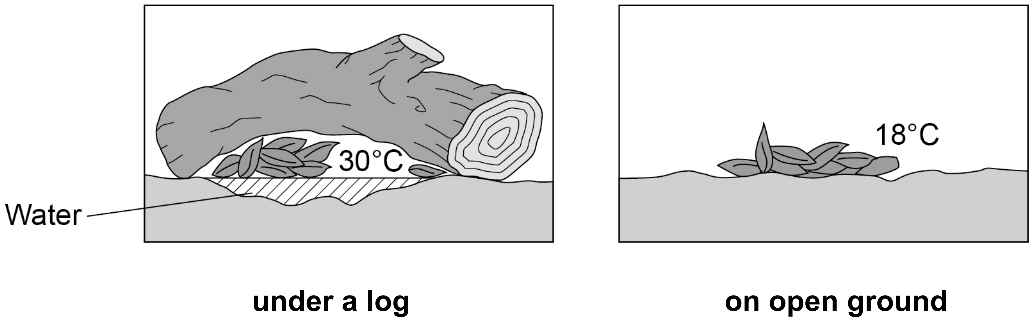
0 5 . 3 Dead leaves are decaying in two places on the forest floor.

Figure 9 shows the two places.

Figure 9



Give **two** reasons why the leaves will decay faster under the log than on the open ground.

[2 marks]

1 ____

____

2 ____

____

0 5 . 4 Describe the process of decay of the leaves.

[2 marks]

____

____

____

____

____

Question 5 continues on the next page

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0 5 . 5 Describe how:

- decay of dead leaves releases substances that new plants can use
- the substances released are then used by new plants to grow.

[6 marks]

12

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0 6 This question is about genetic engineering.

0 6 . 1 Which **two** statements describe stages in genetic engineering?

[2 marks]

Tick (✓) **two** boxes.

A vector is used to insert the gene into the required cells. ☐

Hormones are used to isolate the required gene. ☐

The required gene is cut from a chromosome using an enzyme. ☐

The required gene is transferred to other organisms using gametes. ☐

Tissue culture is used to copy the required gene. ☐

Question 6 continues on the next page

Herbicides are chemicals that can be used to reduce the growth of weeds.

Genetic engineering can produce genetically modified (GM) crop plants that are resistant to herbicides.

Cassava is a crop plant:

- which is an important food source in many countries
- which can be genetically modified to be resistant to a herbicide.

GM cassava is not damaged when sprayed with the herbicide.

Some important amino acids are made in plants from chemical **X**.

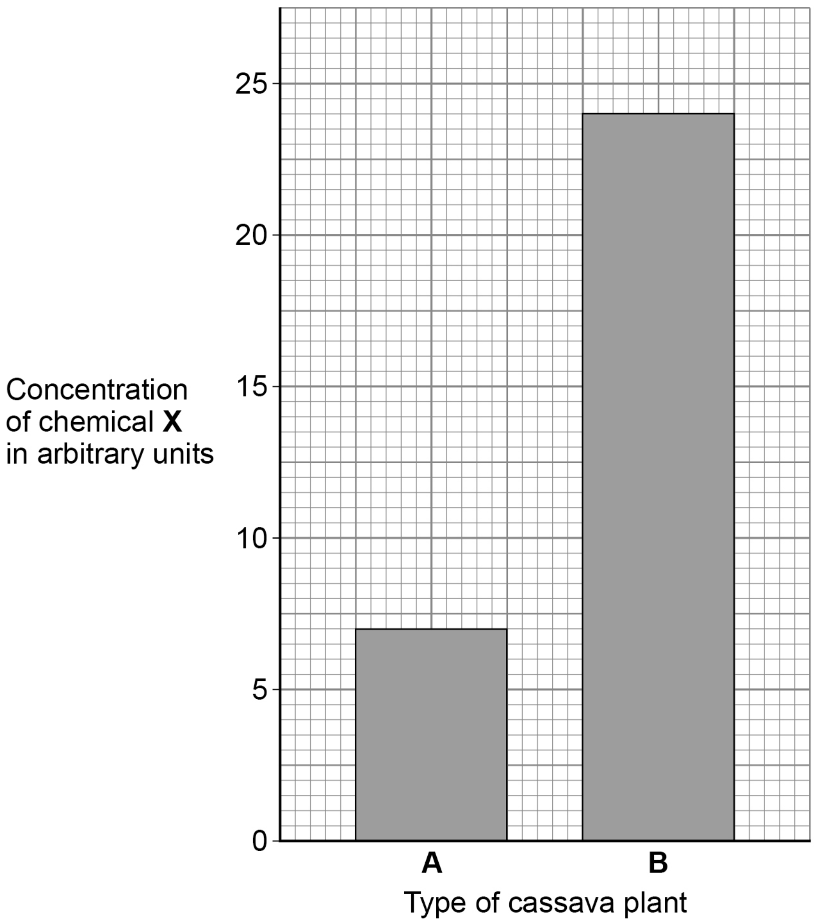
The herbicide works by stopping these amino acids being made from chemical **X**.

Scientists sprayed a normal cassava plant and a GM cassava plant with the herbicide.

The scientists determined the concentration of chemical **X** in the two plants after two weeks.

Figure 10 shows the results.

Figure 10



0 6 . 2 How much greater was the concentration of chemical **X** in plant **B** compared with plant **A**?

Give your answer as a percentage.

Give your answer to 3 significant figures.

[3 marks]

Percentage increase (3 significant figures) ________________________ %

0 6 . 3 'Cassava plant **A** was the genetically modified plant.'

Explain how the results in **Figure 10** support this statement.

[2 marks]

Question 6 continues on the next page

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0 6 . 4 Suggest **three** benefits of growing herbicide resistant GM cassava plants instead of non-GM cassava plants.

[3 marks]

1 ____

____

2 ____

____

3 ____

____

0 6 . 5 Give **two** reasons why some people are concerned about the use of GM crops.

[2 marks]

1 ____

____

2 ____

____

12

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0 7 The heart muscle has its own supply of blood carried by coronary arteries.

In coronary heart disease (CHD) the coronary arteries become narrowed by a build-up of fatty material inside their walls.

0 7 . 1 Suggest **two** ways that a person can reduce their risk of developing CHD.

[2 marks]

1 ____

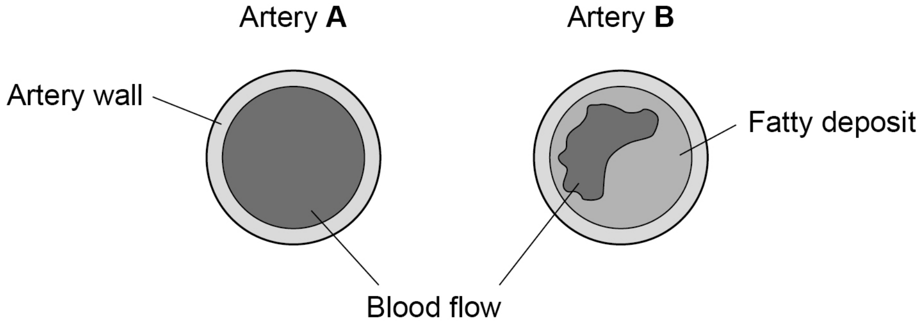
____

2 ____

____

Figure 11 shows coronary arteries from two people.

Figure 11



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0 7 . 2 Explain why the person with coronary artery **B** might become breathless when exercising.

[6 marks]

Question 7 continues on the next page

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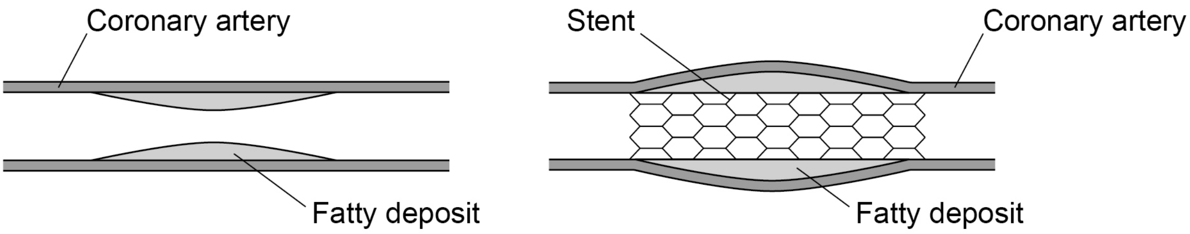
A coronary artery that is blocked with fatty deposits can have a stent inserted to keep it open.

- A stent is a narrow, short tube of flexible metal mesh.
- When the stent is placed in the blocked artery, a small balloon is inflated inside the stent. The balloon is removed and the stent stays in place.

Figure 12 shows a coronary artery before and after a stent is fitted.

Figure 12



0 7 . 3 Suggest why the metal mesh of the stent needs to be flexible.

[1 mark]

Do not write outside the box

0 7 . 4 Suggest why a balloon is used when placing the stent in the coronary artery.

[1 mark]

0 7 . 5 Stents vary in diameter from 2 millimetres to 5 millimetres.

Suggest why stents vary in diameter.

[1 mark]

Question 7 continues on the next page

After the operation to insert a stent some patients are given a blood transfusion.

The blood given to a patient must be of a blood group compatible with the blood group of the patient.

A person's blood group is determined by:

- the antigens on the red blood cells
- the antibodies in the plasma.

Table 3 shows the antigens and antibodies in different blood groups.

Table 3

Blood group	Antigen on red blood cells	Antibody in blood plasma
A	A	anti-b
B	B	anti-a
AB	A and B	none
O	none	anti-a and anti-b

0 7 . 6 A patient has blood group B.

It is **not** safe to give this patient blood from a donor with blood group AB.

Explain why.

Use **Table 3**.

[2 marks]

Red blood cells from blood group O have no antigens on their surfaces.

0 7 . 7 What type of substance is an antigen?

[1 mark]

Do not write outside the box

Scientists have found a way to remove antigens from the red blood cells of blood groups A, B and AB.

0 7 . 8 Explain why it is useful to remove these antigens.

[2 marks]

0 7 . 9 What type of chemical might the scientists use to remove the antigens from the red blood cells?

[1 mark]

17

END OF QUESTIONS

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ANSWER IN THE SPACES PROVIDED

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Question number	Additional page, if required. Write the question numbers in the left-hand margin.

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